

Element L: Presentation of Designer's Recommendations

Even though we were ultimately unable to create a physical prototype due to time and budget constraints, our mathematical and 3D models would be a benefit to anyone approaching a similar problem. With our time on the project coming to a close, we would like to leave the following suggestions for further development/improvement:

- 1) **Use 3D printing or machining to prototype** - Our plan to create the final prototype was to 3D print the chassis and other non-electrical components due to the relatively low costs and time associated with 3D printing, alongside the strength of well-designed 3D printed designs.
 - a) Choosing the prototyping material is also a significant part of prototyping, and we would recommend choosing PLA+, ABS, or TPU. Below we included some of the advantages and disadvantages of each:

Material	Advantages	Disadvantages
PLA+	• Non-toxic • Biodegradable • Highly detailable	• Most brittle • Least durable
ABS	• Strong electrical insulation • Easy to machine and post-process • Relatively strong	• Produces strong fumes • Harder to print with • Prone to warping
TPU	• Extremely flexible and durable • Resistant to most chemicals	• Hardest of three to print

Overall, TPU is generally the best choice for creating prototypes of mechanical designs, but PLA+ is the easiest of the 3 to work with and will likely be strong enough for this project.

- 2) **Add a variable weight system** - As seen in Element E, we determined that the overall weight of the chassis was part of the equation for how much force the

pulling pod can generate. However, increasing the mass will also decrease the acceleration of the pod, so we recommend adding the ability to change the weight of the pod to accommodate a wider range of pulling loads. This could be done by adding circular indents on the top of the pod to place dense disks in to add weight.

- 3) **Focus on the user experience** - One of the best pieces of feedback we received from our expert feedback day was to streamline the user experience. This especially was with respect to the process of attaching the pod to the cart, as it is the one source of possible back pain (bending over) with our solution. One way to make the pod easy to attach and remove, while still having proper control of the cart is to make the hooks/clamp passively open and close. This could look like rubber banding the hooks to the support arms, and then having the geometry of the clamp allow the user to slide the clamp over the edge of the cart.

